# VS-05 Statistical Product Composition

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## VS-05.0 Statistical Product Composition

### Purpose and Scope

This document specifies the complete business layer for the **Statistically composed and composite statistical products** business domain, corresponding to the *VS-05.0 Statistical Product Composition Value Stream*. It also covers 4 governance process types:

- **Design** — how composite products, indicators, and coherence checks are specified and planned (produces metadata)
- **Implementation** — how the composition pipeline is built, configured, and tested (produces the execution process)
- **Execution** — how outputs are operationally integrated, composed, checked, and registered (produces statistical data)
- **Management** — how the domain is monitored, quality-assessed, and improved (produces quality reports)

> **Out of Scope**
>
> Application and Technology layer specifications are out of scope for this document. Re-estimation, weighting, and model-based analysis are out of scope — they belong to VS-04.0 Estimation, Analysis, and Statistical Modeling; VS-05 composes existing validated outputs, it does not re-estimate them. Publishing, distribution, and user services are out of scope — they belong to VS-06.0 Dissemination and Service Delivery.

*A note on numbering:* each `VS-0x.y` below is both a value stream stage (*what value* is added) and a business process (*how* it is done); the numbering maps one-to-one. See the key-concepts box above.

### Position in the Statistical Value Chain

VS-05.0 is the fifth primary value stream — it receives Output Data (validated statistics) from VS-04.0 and brings multiple analytical results together into coherent, high-value statistical products ready for dissemination. It is the "Compose & Approve Products" orchestration stage where separate analyses become finished statistical products.

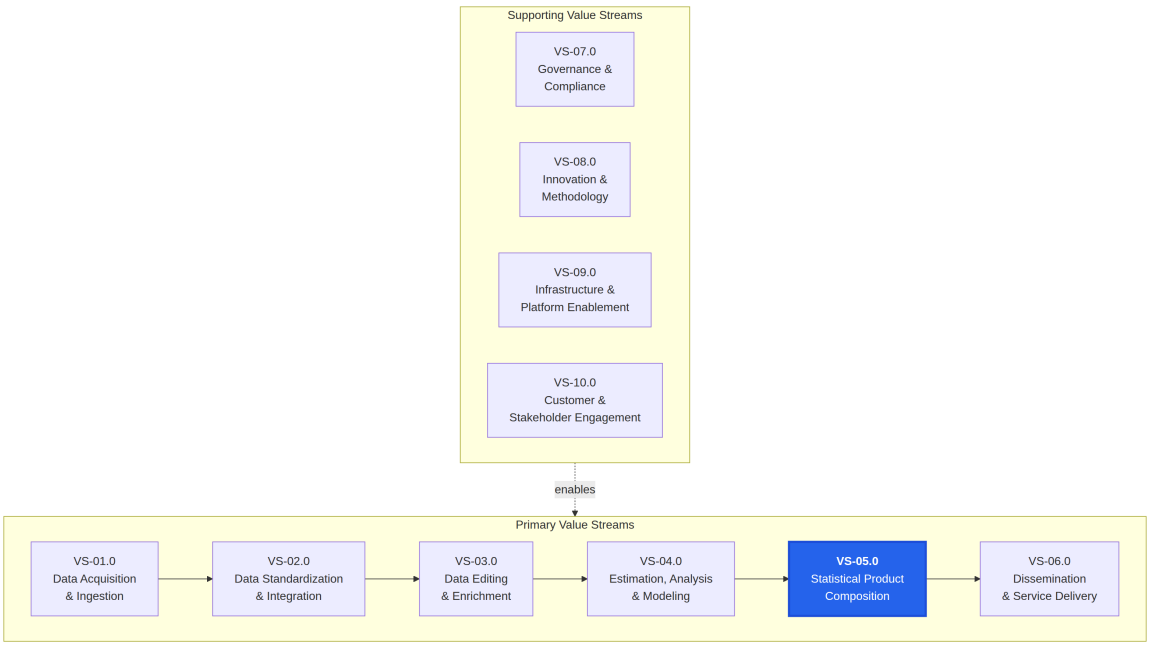



Figure 1: Value Stream Context Map. VS-05.0 is highlighted as the composition stage that turns validated analytical outputs into approved statistical products.

### Steady State Transition

VS-05.0 operates **within Steady State #4 (Output Data / Statistics)**. It receives individual validated analytical outputs (the estimation form of State #4 produced by VS-04.0) and produces **approved, registered composite statistical products** (the final-product form of State #4) — coherent across sources and time, validated against dissemination and confidentiality standards, and ready to hand to dissemination. State #4 is shared between the estimation domain (VS-04) and this composition domain (VS-05).

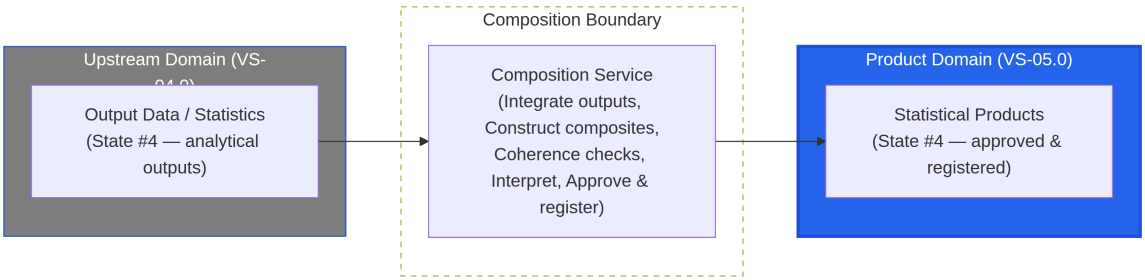



Figure 2: Steady State Transition within State #4: from individual analytical outputs to approved, registered composite statistical products.

Product approval criteria (Examples)*

- Source outputs selected and integrated with their accuracy/methodology metadata preserved
- Composite indicators, indices, and time series constructed per documented specifications
- Internal consistency and coherence with related and prior products verified
- Analytical and interpretive content (commentary, visualizations) produced and reviewed
- Product validated against dissemination and confidentiality standards
- Approved by the designated authority and registered with a unique product identifier
- Full lineage from the product back to its component State #4 outputs and the composition method

The composition discipline — coherence and interpretability

VS-05 does not re-estimate; it **composes** validated outputs into products. The guarantee here is **coherence and interpretability**: composite products are internally consistent, methodologically sound, coherent with related and prior statistics, and accompanied by the metadata and interpretation users need. Every product carries lineage to its component outputs and the composition method, and is **approved against dissemination and confidentiality standards** before it is registered.

Inherited from VS-04.0

VS-05 receives Output Data **plus** the metadata VS-04 attached — methodology, weights, accuracy measures (sampling error, confidence intervals), and the record of the disclosure control already applied. VS-05 **propagates** this accuracy and quality metadata into the composite product (it does not discard or recompute it), uses it to weight or caveat composed figures, and runs coherence checks across the combined outputs. Composition never silently changes a component estimate; any adjustment for coherence is documented and traceable.

### The Four-Process Governance Pattern

### Governing Principles

The following principles directly govern the VS-05.0 business domain:

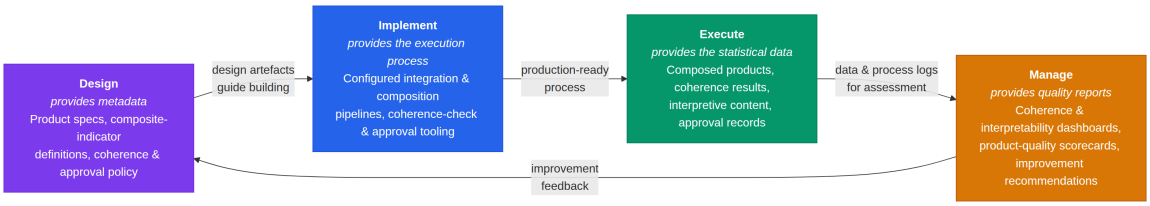



Figure 3: The Four-Process Governance Pattern. Each process type produces a distinct artefact class. The cycle closes when Management findings feed back into Design improvements.

| Principle | Statement | Application to VS-05.0 |
|---|---|---|
| **SAF-P01** | Actively participate in open source to maintain collective control over statistical processes | NSIs should contribute to open source communities for composition, indicator-construction, and reporting/visualization tooling, ensuring collective control over critical product capabilities rather than vendor dependence. |
| **SAF-P07** | Implement loosely coupled processes and systems | The composition domain must be independent from upstream estimation and downstream dissemination. The approved statistical product serves as a decoupling interface — dissemination accesses it without knowledge of how it was composed. |
| **SAF-P08** | Data is shared property | Approved statistical products and their components, once registered, are available to all authorized consumers. Composition should reuse existing validated outputs rather than duplicate analysis. |
| **SAF-P09** | No data without metadata and classification | Every composite product carries metadata: its component outputs and their accuracy, the composition/indicator methodology, coherence-check results, classification, and the approval record. This is what makes products interpretable and trustworthy. |
| **SAF-P10** | Security By Design | Access controls, integrity verification, and a tamper-evident composition and approval record are embedded in the pipeline so that every product is attributable and reproducible, not added as an afterthought. |
| **SAF-P11** | Privacy By Design | Composition is validated against confidentiality standards, and the disclosure risk of composite products (including the risk of combining outputs) is assessed by design, before registration. |

**Cross-cutting principle application**

Several principles cut across all four process types. SAF-P09 (no data without metadata and classification) underpins interpretability: Design defines the product and required metadata, Execute captures component lineage, composition method, and coherence results, and Manage assesses product quality. SAF-P10 (Security By Design) governs the tamper-evident approval/composition record across Execution (Sections 5.1–5.5). SAF-P11 (Privacy By Design) applies especially to approval (Section 5.5), where the composite product's confidentiality and disclosure risk are confirmed before registration. SAF-P07 (loosely coupled) governs both the State #4 intake interface from VS-04.0 and the handover interface to VS-06.0.

### Capability Foundations

VS-05.0 draws on the following SAF capabilities:

| Capability Group | Capabilities | Role in VS-05.0 |
|---|---|---|
| **1.1 Statistical design and build** | 1.1.2 Design statistics product, 1.1.3 Build statistics product | Design the statistical products and composite indicators, and build the composition pipeline |
| **2.1 Data management** | 2.1.1 Data lifecycle management, 2.1.2 Metadata management, 2.1.3 Statistical quality management, 2.1.4 Statistical register management | Manage products, metadata, and quality; assess coherence; integrate across registers and time |
| **3.2 Computerization and automation** | 3.2.1 Application development, 3.2.2 Functional management | Build and maintain integration, indicator-construction, and reporting/visualization engines |
| **3.3 Security and Privacy** | 3.3.1 Privacy management, 3.3.2 Security planning | Validate products against confidentiality standards and assess composite disclosure risk |
| **4.1 Governance** | 4.1.1 Strategy and governance, 4.1.2 Policy and planning, 4.1.4 Improve management, 4.1.5 Accountability | Govern the domain, steward product approval, plan improvements, report on compliance |

### Metadata Perspective

It is considered that metadata is an essential product of the Design phase and a critical concern of every other phase. In this domain, metadata is what makes a composite product interpretable: each product must carry its components, methodology, coherence results, and approval. The table below consolidates the metadata produced across all four process types, covering both data metadata (about the products) and process metadata or paradata (about how the process runs).

| Process Type | Data Metadata Produced | Process Metadata (Paradata) Produced |
|---|---|---|
| **Design** | Product specifications, composite-indicator definitions, coherence criteria, approval & confidentiality policy | Design decisions log, methodology version, review and approval records |
| **Implement** | Integration configurations, indicator-construction parameters, coherence-check parameters, product templates | Build log, test results, coherence-test outcomes, deployment approvals |

| Process Type | Data Metadata Produced | Process Metadata (Paradata) Produced |
|---|---|---|
| **Execute** | Per-product component lineage, composition/indicator method, coherence-check results, interpretive content, approval record | Operator identity, timing, software/version, approval workflow trail |
| **Manage** | Coherence & interpretability assessments, product-quality scores, lineage assessments, trend analyses | Coherence dashboards, product-quality trends, improvement decisions, audit findings |

## Business Roles and Actors

| Role | Process Orientation | Description |
|---|---|---|
| **Upstream Data Provider (VS-04.0)** | External (internal NSI) | Supplies Output Data (State #4) with methodology/accuracy metadata to the composition domain |
| **Standards Body** | External | Maintains product, indicator, and exchange standards the NSI conforms to (e.g., SDMX, index manuals) |
| **Product Manager** | Design | Designs the statistical product portfolio, product specifications, and release content |
| **Indicator/Index Methodologist** | Design | Specifies composite indicators, indices, and time-series construction methods |
| **Coherence Designer** | Design | Designs the consistency and coherence checks across sources and time |
| **Data Engineer** | Implementation | Builds and configures integration, indicator-construction, and reporting engines and the pipeline |
| **System Configurator** | Implementation | Configures composition services, output-source connections, and access controls |
| **Test Manager** | Implementation | Plans and executes testing of the composition pipeline (technical, coherence, reproducibility) |
| **Process Owner** | Management | Accountable for the end-to-end performance of the composition domain |
| **Compliance Officer** | Management | Ensures confidentiality and standards compliance of products, produces audit reports |
| **Quality Manager** | Management | Monitors coherence/interpretability KPIs, produces quality reports, and drives continuous improvement |
| **Integration Specialist** | Execution | Selects and integrates the component outputs into the product structure |
| **Subject-Matter Author** | Execution | Produces analytical and interpretive content, commentary, and visualizations |
| **Approval Authority** | Execution | Validates products against standards and approves them for registration and release |
| **Data Steward** | Execution | Registers approved products and versions, manages metadata entries, and applies access controls |

## Design Processes — "provides metadata"

*Framework alignment: GSBPM Phases 1–2 — see Framework Mappings.*

The design phase produces the **metadata artefacts** that govern all subsequent phases — the product specifications, composite-indicator definitions, coherence checks, and the approval and confidentiality policy. Without well-designed product specs, composition becomes ad-hoc and incoherent, and management has no baseline to measure coherence and interpretability against.

### Design the Statistical Product Portfolio

The starting point is defining what products will be produced. This sub-process specifies each statistical product — its purpose, target users, content, structure, breakdowns, release frequency, and the set of component State #4 outputs it draws on. The team designs how component outputs are selected and integrated (which estimates, which domains, which time periods) and how their accuracy metadata is carried into the product, aligned with GSBPM 2.5 (Design processing and analysis).

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Specify each product: purpose, users, content, structure, breakdowns, frequency | Product Specification | GSBPM 2.1 |
| Define the component outputs each product draws on and selection rules | Component-Selection Specification | GSBPM 2.5 |
| Define how component accuracy/quality metadata is carried into the product | Metadata-Propagation Rules | SAF-P09 |
| Define product classification and versioning identity | Product Metadata Standard | SAF-P09 |

**NSI Example — Labour Market Publication:**

The labour-market quarterly publication is specified to combine the LFS employment-rate estimates (national and NUTS-2), the unemployment series, and administrative job-vacancy figures into a single coherent release, carrying each component's confidence intervals into the published tables.

### Design Composite Indicators and Indices

This sub-process specifies the derived measures the products present — composite indicators, indices, ratios, and time-series aggregates. Each is defined by its formula, component inputs, weighting/aggregation method, base period, and chaining/linking rules for time series, together with the metadata and classification it carries. Where a composite combines outputs of differing accuracy, the design specifies how uncertainty is represented, aligned with GSBPM 2.5.

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Specify composite indicators/indices: formula, inputs, weighting, base period | Composite-Indicator Specification | GSBPM 2.5 |
| Specify time-series construction: chaining, linking, benchmarking | Time-Series Specification | GSBPM 2.5 |
| Define how component uncertainty is represented in composites | Uncertainty-Representation Rules | SAF-P09 |
| Define metadata and classification for each composite | Composite Metadata Standard | SAF-P09 |

### Design Consistency and Coherence Checks

This sub-process designs the checks that ensure integrated products are sound. It specifies internal-consistency checks (totals reconcile to components, breakdowns sum to totals), coherence checks against related statistics (e.g., national accounts, prior products) and across time, and the tolerance thresholds and disposition rules (accept, flag, return for review). The revision policy for products is also designed here, aligned with GSBPM 2.5 and GAMSO Quality Management.

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Define internal-consistency checks (totals, breakdowns reconcile) | Consistency Check Set | GSBPM 2.5 |
| Define coherence checks vs. related statistics and across time | Coherence Check Set | GAMSO QM |
| Define tolerance thresholds and disposition rules | Coherence Tolerance Policy | GSBPM 2.5 |
| Define product revision policy | Revision Policy | GAMSO QM |

#### Design Analytical and Interpretive Content

This sub-process designs the human-facing content that makes products interpretable. It specifies the analytical narrative and commentary structure, the visualizations and tables, the key messages and their evidence, and the editorial/quality-review workflow for interpretive content. Accessibility and plain-language standards are designed in, aligned with GSBPM 2.5.

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Specify analytical narrative and commentary structure | Content Template | GSBPM 2.5 |
| Specify visualizations and tables | Visualization Specification | GSBPM 2.5 |
| Define key messages and evidence requirements | Key-Message Guidelines | GSBPM 2.5 |
| Define editorial and quality-review workflow for content | Editorial Review Policy | GAMSO QM |

#### Design Approval, Confidentiality, and Versioning Framework

The final design sub-process ensures products are properly approved, protected, and traceable. It defines the approval workflow and authority, the confidentiality/disclosure standards a product must meet before registration (including the risk of combining outputs), the product-versioning policy (identity, revision triggers, immutability, retention), the lineage requirements from each product to its component outputs, and the role-based access controls in line with SAF-P10.

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Define approval workflow and authority | Approval Policy | SAF 4.1.5 |
| Define confidentiality/disclosure standards for products (incl. combination risk) | Confidentiality Standard | SAF-P11 |
| Define product-versioning and lineage requirements | Versioning & Lineage Policy | CSDA Traceability |
| Design role-based access control for products | Access Control Design | SAF-P10 |

## Implementation Processes — "provides the execution process"

*Framework alignment: GSBPM Phase 3 (Build) — see Framework Mappings.*

The implementation phase takes the metadata artefacts from Design and **builds, configures, and tests** the production-ready composition pipeline. Nothing runs operationally until this phase is complete.

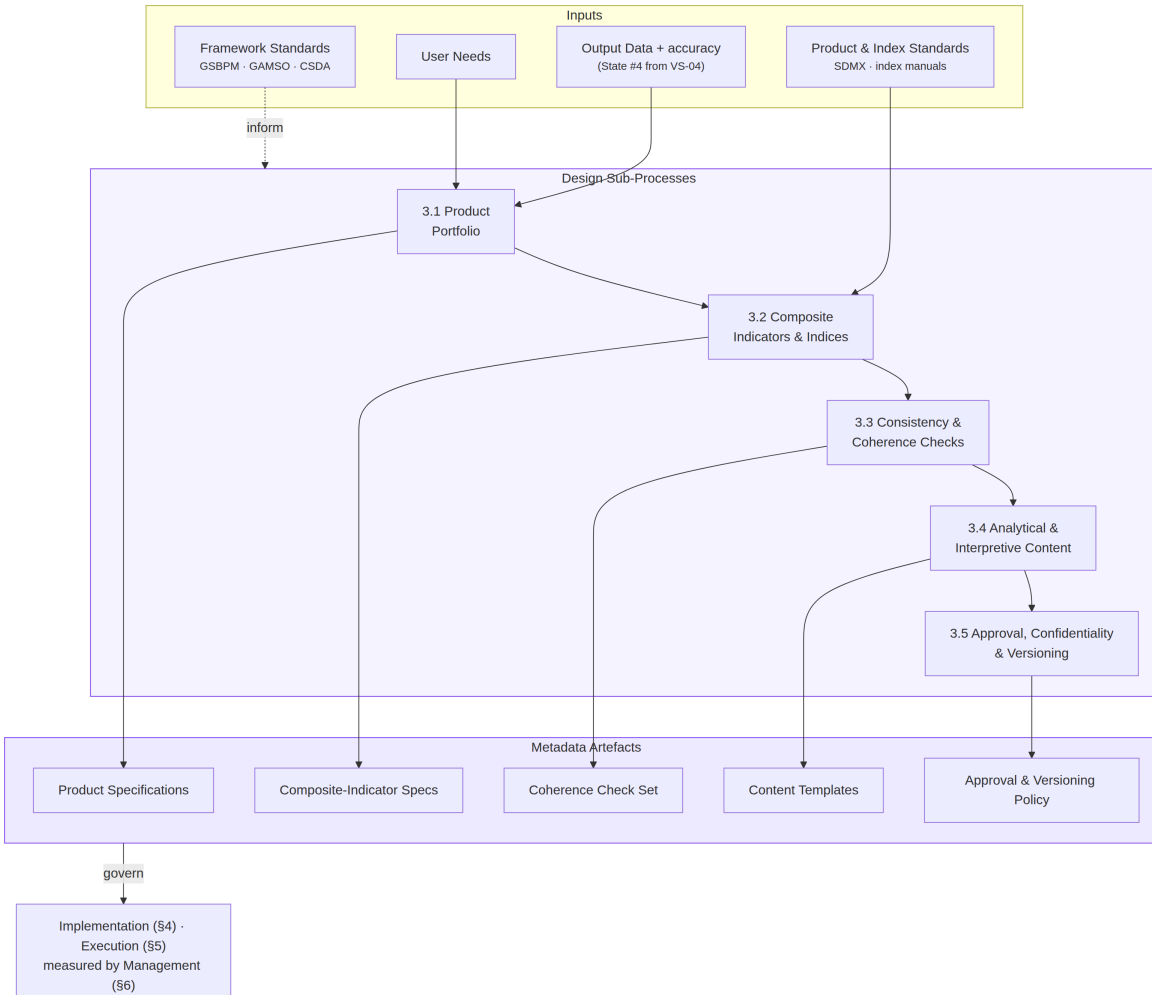



Figure 4: Design Process Decomposition. Output data with accuracy, user needs, and product standards flow into five design sub-processes, producing metadata artefacts that govern Implementation, Execution, and Management.

### Build Integration and Composition Engines

This sub-process turns the product and integration designs into operational tools. The integration engine is built or configured to select and combine the designed component outputs into the product structure, carrying their accuracy/quality metadata. The composition engine is built to assemble products from their components per the specifications. Output-source connections are configured to the State #4 catalog with version pinning so the exact component versions used are recorded. All composition logic is version-controlled in line with SAF-P10, following GSBPM 3.1–3.2.

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Build/configure the integration engine (select and combine component outputs) | Configured integration engine | GSBPM 3.2 |
| Build/configure the composition engine (assemble products from components) | Configured composition engine | GSBPM 3.2 |
| Configure pinned connections to the State #4 output catalog | Output-source integration | CSDA Metadata Mgmt |
| Version-control all composition logic | Versioned composition repository | SAF-P10 |

**NSI Example — Business Statistics Product:**

The SBS composite product engine is configured to integrate the register-based aggregates and the survey-based estimates for the same NACE × region domains, pinning the exact `OD-BUS` output versions and carrying their accuracy metadata into the combined tables.

### Build Indicator-Construction and Time-Series Engines

The designed composite-indicator and time-series specifications are implemented. Indicator/index calculators are built from the designed formulae and weighting, time-series engines are configured for chaining, linking, and benchmarking, and uncertainty representation is implemented so composites carry an appropriate measure of accuracy. Every composite output is produced with its method and inputs attached, following GSBPM 3.2.

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Build indicator/index calculators from designed formulae | Indicator-construction engine | GSBPM 3.2 |
| Configure time-series chaining, linking, and benchmarking | Time-series engine | GSBPM 3.2 |
| Implement uncertainty representation for composites | Composite-accuracy component | SAF-P09 |
| Attach method and inputs to every composite | Composite metadata binding | SAF-P09 |

### Build Coherence-Check, Content, and Approval Tooling

This sub-process implements the consistency/coherence checks, the content-production tooling, and the approval/versioning store. Coherence tooling implements the designed internal-consistency and cross-source/time checks with their tolerances. Content tooling supports the analytical narrative, tables, and visualizations per the templates, with the editorial review workflow. The approval-and-versioning store is built so each approved product version is immutable, identified, reproducible from its components plus method, and accompanied by its lineage and approval record. Event notifications are configured to alert downstream consumers (VS-06.0) when a product is approved and registered.

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Implement coherence/consistency checks with tolerances | Coherence tooling | GSBPM 3.2 |
| Build content/visualization tooling with editorial workflow | Content tooling | GSBPM 3.2 |
| Build the approval-and-versioning store (immutable, identified, reproducible) | Approval & versioning store | CSDA Traceability |
| Configure "Statistical Product Registered" event for downstream consumers | Event publication | GSBPM 3.2 |

### Test and Validate the Pipeline

Before the pipeline goes into production, it must be rigorously verified. Technical testing covers all components — integration, composition, indicator/time-series engines, coherence checks, content tooling, and the approval store (GSBPM 3.5). Integration testing validates the end-to-end flow from output intake through composition, coherence checks, content, approval, to registration. A **coherence test** confirms composites reconcile to components and cohere with prior products; a **reproducibility test** confirms a product regenerates identically from its component versions plus the composition method. The phase concludes with finalization of the production system and documentation of operational procedures (GSBPM 3.6–3.7).

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Technical testing of all individual components | Unit test reports | GSBPM 3.5 |
| Integration testing: end-to-end flow from output intake to registration | Integration test report | GSBPM 3.5 |
| Coherence test (composites reconcile to components and prior products) | Coherence test report | GSBPM 3.6 |
| Reproducibility test (regenerate a product from components + method) | Reproducibility report | GSBPM 3.6 |
| Finalize production system and document operational procedures | Operational Procedures Manual, Production-Ready Pipeline | GSBPM 3.7 |

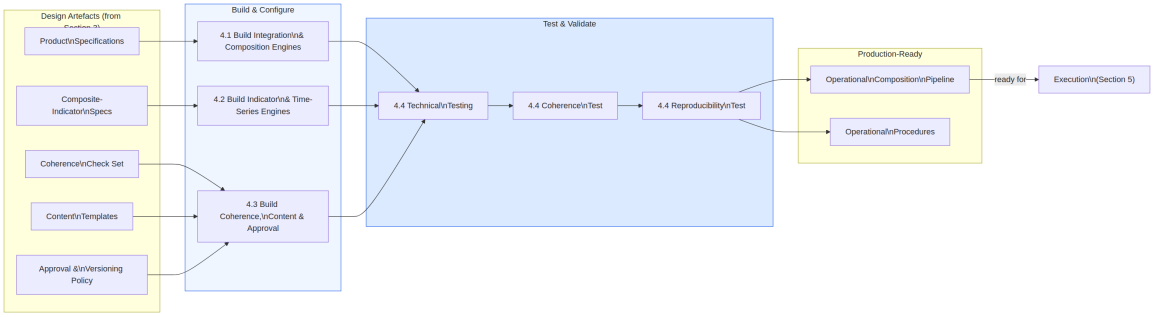



Figure 5: Implementation Pipeline from design artefacts through build/configure/test to production-ready system.

## Execution Processes — "provides the statistical data"

*Framework alignment: the later GSBPM phases for this domain — see Framework Mappings.*

The execution phase runs the **designed and implemented pipeline** operationally for each production cycle. This corresponds to the five VS-05 value stream stages. In a mature steady state, much of the work has been pre-designed (Section 3) and pre-built (Section 4) — execution focuses on running the cycle, composing products, checking coherence, and capturing component lineage and approval so that every product is coherent, interpretable, and reproducible.

### Select and Integrate Source Datasets

**Value created:** *The right validated outputs brought together as the basis for a coherent product.*

When an "Output Data Registered" event is received from VS-04.0, execution begins by selecting and integrating the component outputs for the product. Per-cycle execution applies the designed selection rules to pick the correct State #4 output versions (by domain, period, and accuracy), integrates them into the product structure, and carries their accuracy/methodology metadata forward. Components from multiple domains, surveys, or time periods are combined per the specification, aligned with GSBPM 7.1/7.2.

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Select the correct component output versions per the specification | Selected components | GSBPM 7.1 |
| Integrate components into the product structure | Integrated product base | GSBPM 7.2 |
| Carry component accuracy/methodology metadata into the product | Propagated metadata | SAF-P09 |
| Record component lineage (which output versions were used) | Component lineage | CSDA Traceability |

**NSI Example — Labour Market Publication:**

For Q3-2026 the team selects the validated `OD-LFS-2026Q3` employment and unemployment outputs (national and NUTS-2) and the latest administrative vacancy output, pinning their exact versions and carrying their confidence intervals into the integrated product base.

### Construct Composite Indicators or Indices

**Value created:** *Derived measures — indices, ratios, time series — that summarize and add meaning.*

This sub-process computes the designed composite indicators, indices, and time-series aggregates from the integrated components, aligned with GSBPM 7.2. Indicators and indices are calculated per the designed formulae and weighting; time series are chained, linked, and benchmarked per the specification; and each composite carries an appropriate uncertainty measure derived from its components. No component estimate is re-estimated — composites are built from the validated outputs as received.

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Calculate composite indicators/indices per designed formulae | Composite indicators | GSBPM 7.2 |
| Construct time series (chain, link, benchmark) | Composite time series | GSBPM 7.2 |
| Derive composite uncertainty from component accuracy | Composite accuracy measures | SAF-P09 |
| Record composition method and inputs as lineage | Composite lineage | SAF-P09 |

**NSI Example — CPI Headline Index:**

The headline CPI is composed from the validated item indices using the designed weighting and chaining; the all-items index and its main divisions are constructed, with each carrying its method and the period weights used.

### Perform Consistency and Coherence Checks

**Value created:** *Confidence that the product is internally consistent and coheres with the wider statistical picture.*

This sub-process runs the designed checks, aligned with GSBPM 6.2 (Validate outputs). Internal-consistency checks confirm totals reconcile to components and breakdowns sum to totals; coherence checks compare the product against related statistics and prior releases within the designed tolerances. Discrepancies are dispositioned per policy — accepted, flagged, or returned for review — and explained where significant. No silent adjustment is made: any reconciliation is documented and traceable.

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Run internal-consistency checks (totals, breakdowns reconcile) | Consistency results | GSBPM 6.2 |
| Run coherence checks vs. related and prior statistics | Coherence results | GSBPM 6.2 |
| Disposition discrepancies (accept / flag / return) and explain | Discrepancy dispositions | GSBPM 6.2 |
| Log all check results to the product audit trail | Audit trail entries | SAF-P10 |

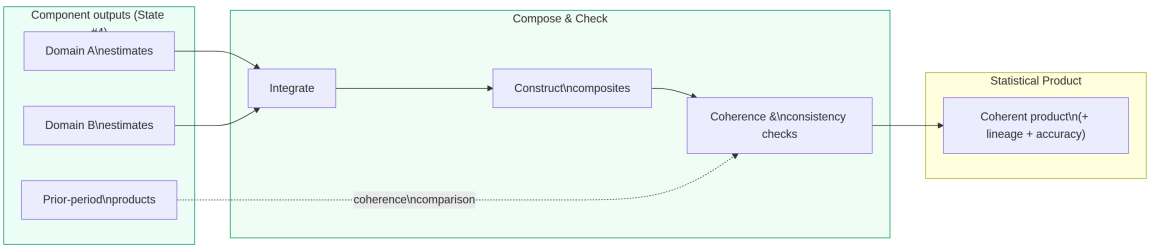



Figure 6: Composition and Coherence Architecture. Component outputs are integrated and composed into indicators, then checked for internal consistency and coherence against prior products before becoming a coherent product carrying its lineage and accuracy.

**NSI Example — Business Statistics Product:**

The composite SBS product is checked: NACE-class turnover sums to the published total, the survey- and register-based components agree within tolerance for overlapping domains, and the series is coherent with the prior year; one domain exceeds tolerance and is returned to VS-04 for review rather than silently adjusted.

### Generate Analytical and Interpretive Outputs

**Value created:** *Products users can understand — narrative, key messages, and visualizations grounded in the data.*

This sub-process produces the human-facing content, aligned with GSBPM 6.3 (Interpret and explain outputs). The analytical narrative and commentary are written per the templates, key messages are stated with their supporting evidence, and tables and visualizations are produced. Content passes the editorial

and quality-review workflow to ensure it is accurate, plain-language, accessible, and consistent with the data and its caveats (including the accuracy carried from VS-04).

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Write analytical narrative and commentary per templates | Product narrative | GSBPM 6.3 |
| State key messages with supporting evidence | Key messages | GSBPM 6.3 |
| Produce tables and visualizations | Visualizations & tables | GSBPM 6.3 |
| Run editorial and quality review of content | Reviewed content | GAMSO QM |

**NSI Example — Labour Market Publication:**

The quarterly commentary explains the 0.3pp rise in the employment rate, notes it is within the confidence interval, and presents a regional map and a time-series chart; the editorial review confirms the wording does not overstate a movement that is not statistically significant.

#### Approve and Register Statistical Products

**Value created:** *A validated, approved, registered product ready for dissemination.*

The final execution sub-process validates, approves, and registers the product, aligned with GSBPM 6.5 (Finalise outputs) and 7.1. The product is validated against the dissemination and confidentiality standards (including the disclosure risk arising from combining outputs), submitted to the designated approval authority, and — on approval — a persistent immutable product version is created with a unique identifier and full lineage (components, method, coherence results, approval record), registered per SAF-P09, with access permissions applied per SAF-P10. The sub-process concludes by publishing a "Statistical Product Registered" event, notifying downstream VS-06.0 that an approved product is available for dissemination.

**Output: Approved Statistical Product (Steady State #4)** — the official handover artefact to the dissemination domain.

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Validate the product against dissemination and confidentiality standards | Validation result | GSBPM 6.5 |
| Submit to and obtain sign-off from the approval authority | Approval record | SAF 4.1.5 |
| Create an immutable, identified product version with full lineage | Versioned product | CSDA Traceability |
| Register metadata and publish "Statistical Product Registered" event to notify VS-06.0 | Catalog registration, event | SAF-P07 |

## Management Processes — "provides quality reports"

*Framework alignment: GSBPM overarching Quality/Metadata Management — see Framework Mappings.*

The management phase runs **alongside and after** execution. It monitors the composition domain's performance, assesses the coherence and interpretability of products, ensures every product is approved and reproducible, and feeds improvement recommendations back into the design phase — closing the governance cycle.

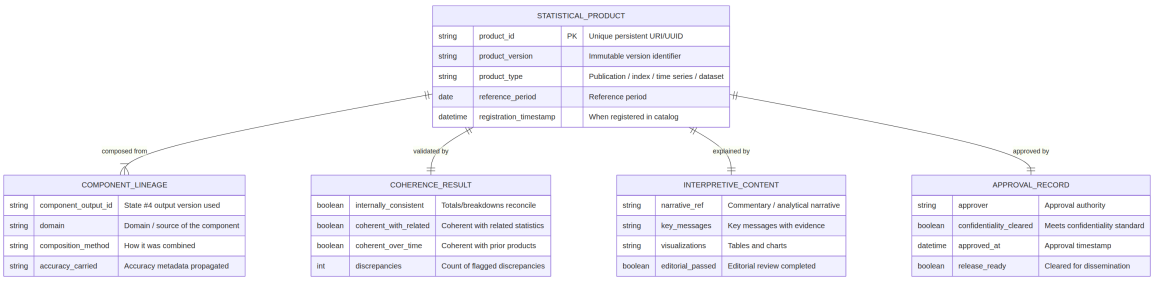



Figure 7: Statistical Product Artefact Structure. Reading guide: each box is a type of record with its fields; a connecting line means the records are linked. Each product version is linked to its component lineage, coherence results, interpretive content, and approval record, all traceable to the State #4 outputs it was composed from.

### Monitor Process Performance

Operational performance of the composition domain must be tracked systematically across production cycles. The process owner and quality manager collect and aggregate key performance indicators, monitor throughput against the release calendar, and track paradata — which component versions and methods were used, when, by whom, and how many approval iterations occurred. This information feeds process performance dashboards that give the process owner a real-time view of domain health, aligned with the GAMSO Quality Management capability.

**Key Performance Indicators (KPIs):**

| KPI | Description | Measurement |
|---|---|---|
| On-time product rate | Products composed within the release-calendar window | Per product, per cycle |
| Coherence-check pass rate | Products passing coherence checks on first run | Per product, per cycle |
| Approval-iteration count | Number of approval cycles before sign-off | Per product, per cycle |
| Component reuse rate | Share of products reusing existing validated outputs | Per cycle |
| Reproducibility rate | Products regenerable from components + method | Per cycle |
| Processing time | Duration from output intake to product registration | Per product, per cycle |

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Collect and aggregate KPIs per production cycle | Process Performance Dashboard | GAMSO QM |
| Monitor throughput against the release calendar | Throughput Report | GAMSO QM |
| Track paradata: component versions, methods, approval iterations | Paradata Records | GAMSO QM |
| Generate process performance dashboards | Dashboard for Process Owner | GAMSO QM |

### Assess Product Quality

Beyond process performance, the quality of the products themselves must be evaluated against the criteria established in design. Each product is assessed across multiple quality dimensions: coherence (internal

consistency and consistency with related and prior statistics), interpretability (clarity and accuracy of narrative and visualizations), completeness (all designed content present), and accuracy propagation (component accuracy correctly carried). Results are compared against the designed quality criteria from Section 3.5. Quality indicators are produced per product and per cycle, and trends are tracked to detect recurring coherence issues or interpretive weaknesses. This aligns with the CSDA Data Quality capability and SAF Capability 2.1.3.

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Evaluate coherence (internal and with related/prior statistics) | Coherence Report (per cycle) | SAF 2.1.3 |
| Assess interpretability of narrative and visualizations | Interpretability Review | CSDA Data Quality |
| Check completeness and correct accuracy propagation | Product Completeness Report | SAF 2.1.3 |
| Identify quality trends across products and cycles | Quality Trend Analysis | SAF 2.1.3 |

#### Manage Product and Version Lifecycle

The product portfolio and the versions it produces must remain current, complete, and well-governed. This sub-process governs the lifecycle of product specifications, composite-indicator definitions, and content templates — versioning them, managing transitions (e.g., an index rebasing), and retiring obsolete products. It verifies metadata completeness — that every product carries full component lineage, composition method, coherence results, and approval as an SAF-P09 compliance check — and governs the product-version store: retention, immutability, and reconciliation that catalog entries match stored versions. Metadata lineage from outputs through composition to products is tracked.

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Govern product, indicator, and template lifecycle and versions | Product Version Register | SAF-P09 |
| Verify metadata completeness (component lineage, method, coherence, approval) | Metadata Quality Report | SAF-P09 |
| Govern the product-version store: retention, immutability, reconciliation | Version Store Audit | CSDA Traceability |
| Track metadata lineage from outputs to products | Metadata Lineage Audit | CSDA Traceability |

#### Audit and Compliance Reporting

This sub-process produces the evidence that the composition domain operates within methodological, legal, and organizational requirements. Reproducibility-audit reports demonstrate that every product can be regenerated from its component versions plus the composition method — supporting transparency and accountability. Approval-and-confidentiality reporting confirms that every registered product was approved by the designated authority and meets the confidentiality standard, including the disclosure risk of combining outputs per SAF-P11. Standards compliance reporting confirms conformance to product and exchange standards (e.g., SDMX), and security compliance reporting verifies access controls and the integrity of the approval record. These reports satisfy the SAF Capability 4.1.5 (Accountability) obligation and feed into VS-07.0 Governance and Compliance.

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Produce reproducibility-audit reports: every product regenerable | Reproducibility Audit Summary | SAF 4.1.5 |
| Report on approval and confidentiality compliance of products | Approval & Confidentiality Report | SAF-P11 |
| Report on standards compliance (products, SDMX exchange) | Standards Compliance Report | SAF 4.1.5 |
| Report on security compliance: access controls, approval-record integrity | Compliance Report (per cycle) | SAF-P10 |

#### Continuous Improvement

The final management sub-process closes the governance loop by analyzing performance and quality data to identify and drive improvements. Quality trends across production cycles are examined: are coherence issues falling, are approval iterations shrinking, is component reuse rising, is interpretability improving? Improvement opportunities are identified — refining composite-indicator methods, improving coherence checks that generate false positives, streamlining the approval workflow, enhancing visualizations, and increasing reuse of validated outputs. These improvement recommendations are fed back to the design processes (Section 3), prioritized based on impact and feasibility, and tracked through subsequent design-implement-execute cycles. This is the SAF Capability 4.1.4 (Improve management) in action, and it connects to the GAMSO concept of transferring mature capabilities from innovation into production.

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Analyze quality trends across production cycles | Improvement Recommendations Report | SAF 4.1.4 |
| Identify improvement opportunities: indicator methods, coherence checks, approval, visualization, reuse | Updated Design Inputs | SAF 4.1.4 |
| Feed recommendations back to design processes (Section 3) | Design feedback (closing the governance loop) | SAF 4.1.4 |
| Prioritize improvements based on impact and feasibility | Improvement Tracking Log | GAMSO Capability Mgmt |
| Track implementation of accepted improvements | Improvement Tracking Log | GAMSO Capability Mgmt |

## Business Services and Interfaces

The VS-05.0 business domain exposes the following services:

| Service | Direction | Consumers | Mapped Processes |
|---|---|---|---|
| **Integration and Composition Service** | Internal (VS-04.0 → NSI) | Output data consumers, composition teams | 5.1 Select and Integrate Source Datasets |
| **Indicator and Index Service** | Internal | Composition teams, methodologists | 5.2 Construct Composite Indicators or Indices |
| **Coherence Validation Service** | Internal | Quality managers, subject-matter authors | 5.3 Perform Consistency and Coherence Checks |

| Service | Direction | Consumers | Mapped Processes |
|---|---|---|---|
| **Analytical Content Service** | Internal | Authors, editors, downstream dissemination | 5.4 Generate Analytical and Interpretive Outputs |
| **Product Provisioning Service** | Internal (NSI → VS-06.0) | Dissemination teams | 5.5 Approve and Register Statistical Products |
| **Product Catalog Service** | Internal | All data consumers across the NSI | 6.3 Manage Product & Version Lifecycle |
| **Design and Methodology Service** | Internal | Product managers, indicator methodologists | 3.1–3.5 All design processes |

**Interface specifications:**

- **Product Provisioning Service** interface: event-driven notification ("Statistical Product Registered") + catalog query API for downstream consumers to discover and access approved products, their lineage, and accuracy
- **Indicator and Index Service** interface: composite-construction API and a methodology/parameters query API
- **Product Catalog Service** interface: search/browse API for products, component lineage, coherence results, and approval records

### Boundary Interface Specifications

The domain is loosely coupled (SAF-P07) to its neighbors through two steady-state interfaces. Specifying each boundary explicitly — the handover artefact, the metadata it must carry, the event that triggers it, and where responsibility transfers — makes the loose-coupling contract concrete and testable.

| Boundary | Handover Artefact | Required Metadata | Trigger Event | Responsibility Transfer |
|---|---|---|---|---|
| **VS-04.0 → VS-05.0** (intake) | Output Data / Statistics (State #4) | Methodology and parameters, weights, accuracy measures, model diagnostics, applied-SDC record, output-version identifier, lineage to processed source, BIV classification | "Output Data Registered" event | VS-04.0 retains custody of the output version; VS-05.0 assumes responsibility for composing products on event receipt |

| Boundary | Handover Artefact | Required Metadata | Trigger Event | Responsibility Transfer |
|---|---|---|---|---|
| **VS-05.0 → VS-06.0** (handover) | Approved Statistical Product (State #4) | Component lineage, composition/indicator method, propagated accuracy, coherence results, interpretive content, approval record, confidentiality status, product-version identifier, BIV classification | "Statistical Product Registered" event | VS-05.0 retains custody of the product version; VS-06.0 assumes responsibility for dissemination and service delivery on event receipt |

Each interface exchanges a registered, immutable steady-state artefact plus its metadata — never internal working data — so neither side needs knowledge of the other's internal processing.

## Business Objects and Data Flow

## Business Events

## Framework Mappings

This reference table maps each process group of this document to the international frameworks (GSBPM, GAMSO, CSDA) and the SAF capability model. It is intended for architects and auditors; the narrative sections above do not depend on it.

| Process group | GSBPM | GAMSO | CSDA | SAF Capabilities |
|---|---|---|---|---|
| **Design** | Phase 1 (Specify Needs), Phase 2 (Design: 2.1–2.5) | Production (design activities), Capability Management | Data Governance, Metadata Management, Data Quality | 1.1.2 Design statistics product, 2.1.2 Metadata management, 2.1.3 Statistical quality management |
| **Implementation** | Phase 3 (Build: 3.1–3.7) | Capability Management | Data Processing / Dissemination preparation (setup), Data Governance (configuration) | 1.1.3 Build statistics product, 3.2.1 Application development, 3.2.2 Functional management |

| Process group | GSBPM | GAMSO | CSDA | SAF Capabilities |
|---|---|---|---|---|
| **Execution** | Phase 6 (Analyze: 6.2 Validate outputs, 6.3 Interpret & explain, 6.5 Finalise outputs) + Phase 7 (7.1 Update output systems, 7.2 Produce dissemination products) | Production (execution) | Data Analysis / Dissemination preparation (execution) | 2.1.2 Metadata management, 2.1.3 Statistical quality management |
| **Management** | Overarching processes (Quality Management, Metadata Management) | Corporate Support (Quality Management, Data Management, Metadata Management), Capability Management (improvement) | Data Quality, Data Governance, Traceability | 2.1.3 Statistical quality management, 4.1.4 Improve management, 4.1.5 Accountability |

## Applying This to Your NSI

### Maturity Assessment Checklist

Use this checklist to assess your NSI's maturity across all four process types — not just execution.

**Design Maturity**

- ☐ Do we have documented product specifications with their component outputs and selection rules?
- ☐ Do we have formal composite-indicator and time-series construction specifications?
- ☐ Do we have documented internal-consistency and coherence checks with tolerances?
- ☐ Do we have content templates and an editorial-review policy for interpretive content?
- ☐ Do we have an approval, confidentiality, and product-versioning policy defined before composition begins?

**Implementation Maturity**

- ☐ Are our composition engines built from design specifications (not ad-hoc spreadsheets)?
- ☐ Do composites carry an uncertainty measure derived from their components?
- ☐ Can a product be reproduced identically from its component versions plus the composition method?
- ☐ Is the approval workflow and versioning store implemented and tamper-evident?
- ☐ Do we have a formal testing process (technical, coherence, reproducibility)?

**Execution Maturity**

- ☐ Do we select pinned component output versions and carry their accuracy metadata forward?
- ☐ Do we construct composites without re-estimating component outputs?
- ☐ Do we run internal-consistency and coherence checks and explain discrepancies?
- ☐ Do we produce reviewed interpretive content consistent with the data and its caveats?
- ☐ Do we approve and register every product as an immutable version with full lineage?

**Management Maturity**

- ☐ Do we track KPIs (coherence pass rate, approval iterations, reproducibility) across cycles?
- ☐ Do we assess coherence and interpretability of products?
- ☐ Do we govern product/version lifecycles and the product-version store?
- ☐ Do we produce reproducibility-audit, approval/confidentiality, and standards-compliance reports?

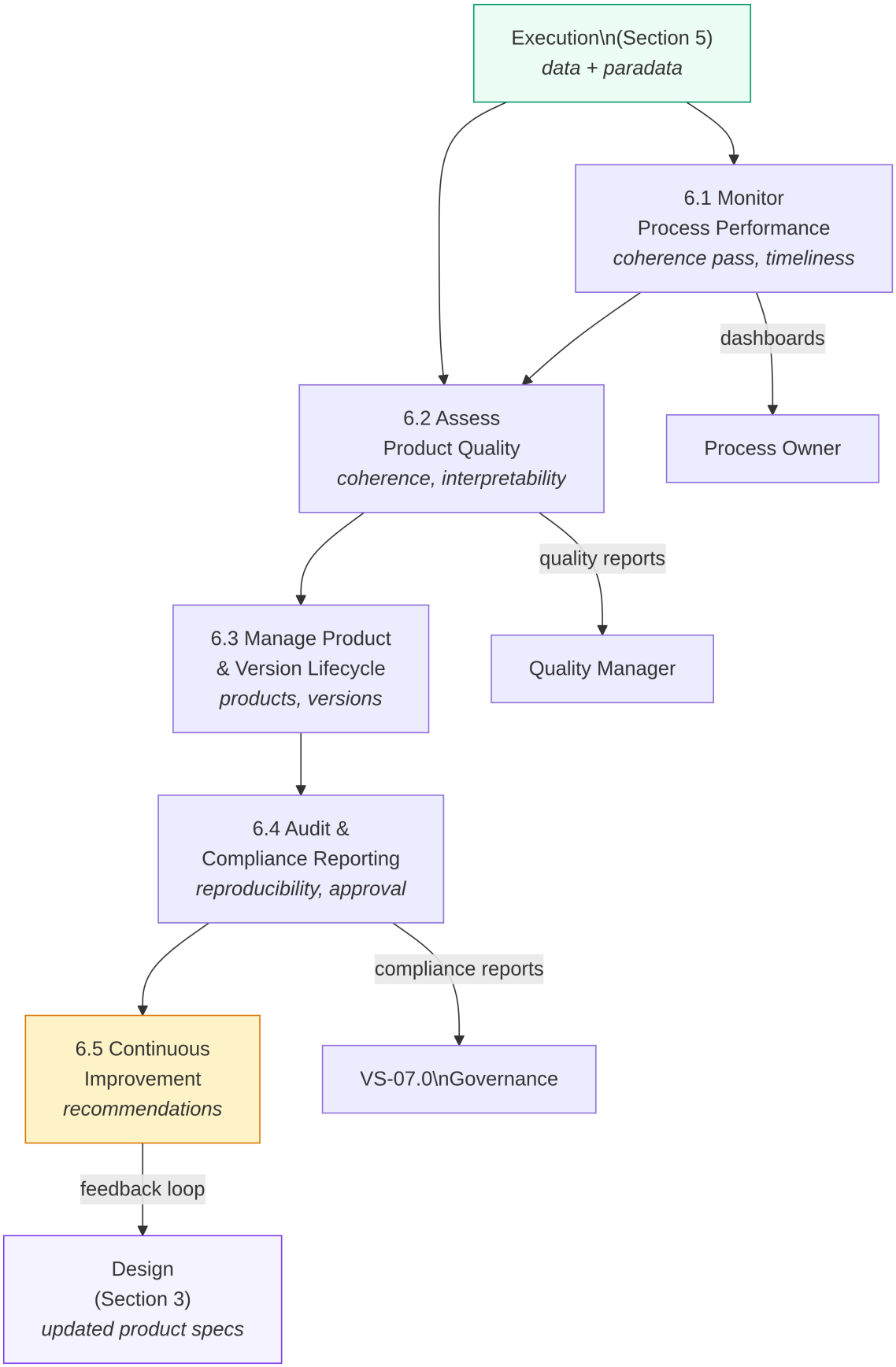



Figure 8: Management Feedback Loop. Execution data flows through monitoring, quality assessment, product/version management, and audit processes. Continuous improvement feeds recommendations back to Design, closing the governance cycle.

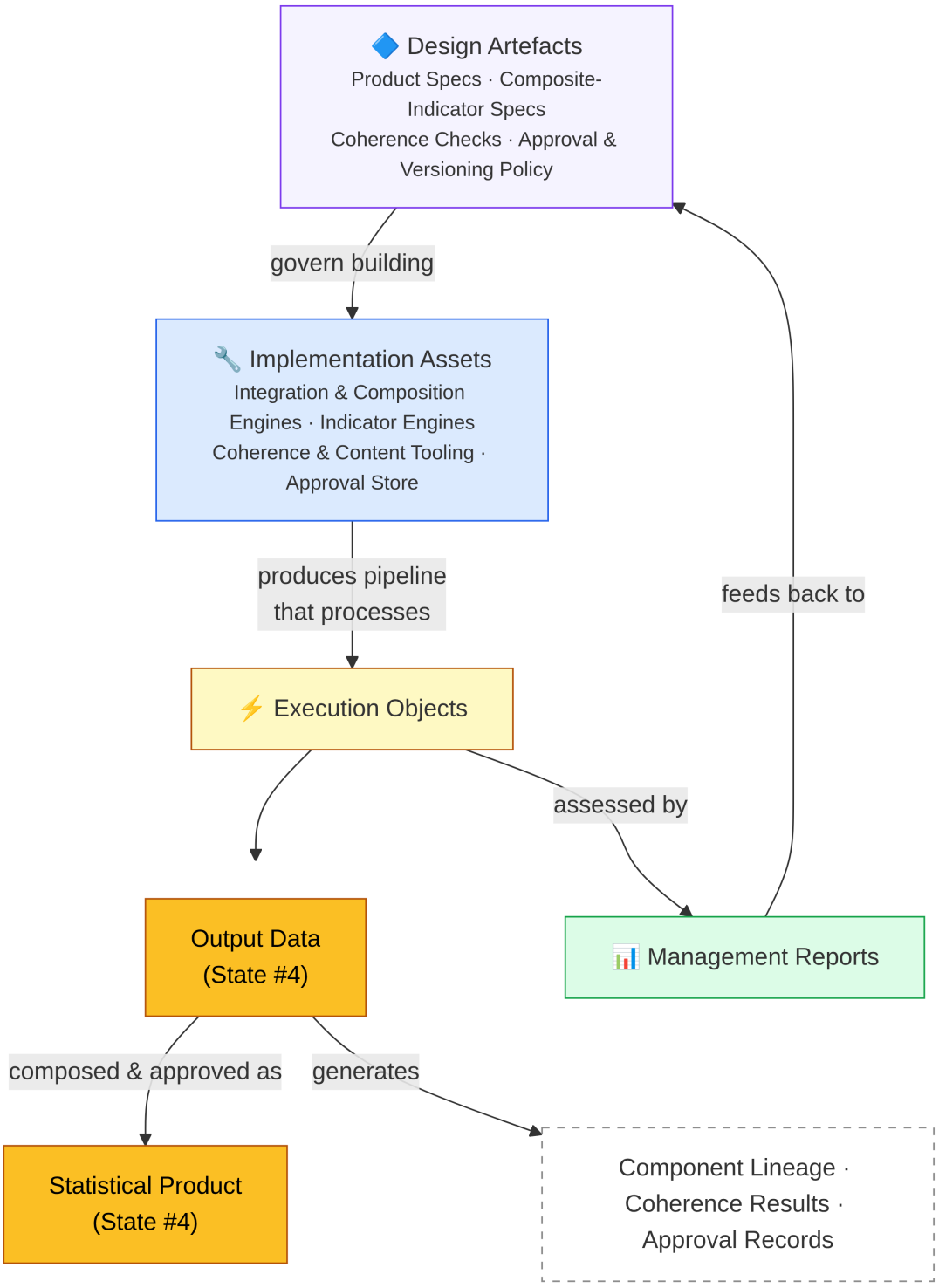



Figure 9: Business Object Lifecycle across all four process types. Design objects govern implementation, which produces the pipeline that composes execution objects into products, which are assessed by management objects, which feed back to design.




Figure 10: Event-Driven Choreography between the upstream domain (VS-04.0) and NSI teams, showing events across the Design → Implement → Execute → Manage cycle. The management phase closes the loop by feeding improvements back to design.

☐ Do we have a formal improvement process that feeds back into design?

#### End-to-End Scenarios

Each scenario walks through **all four process types**, demonstrating how the governance pattern works in practice.

#### Scenario A: Labour Market Quarterly Publication

**Design Phase:** Product managers specify the quarterly labour-market publication: employment and unemployment rates (national and NUTS-2), a vacancy indicator, narrative, and a regional visualization, drawing on the validated LFS and administrative outputs. Coherence checks against prior quarters and the approval workflow are designed.

**Implementation Phase:** Engineers configure the integration engine (pinned LFS/administrative outputs), the time-series engine for the headline series, coherence checks vs. prior quarters, and the content/approval tooling. Coherence and reproducibility tests pass on historical data.

**Execution Phase:** For Q3-2026, an Output Data Registered event arrives. The team integrates the pinned `OD-LFS-2026Q3` outputs (carrying their CIs), constructs the seasonally-relevant series, and runs coherence checks (all reconcile; coherent with Q2). The commentary explains the 0.3pp employment-rate rise and notes its significance; visualizations are produced and editorially reviewed. The product is approved and registered as an immutable version with full lineage; VS-06.0 is notified.

**Management Phase:** Coherence-check pass rate is 100%, approval completed in one iteration, all component CIs correctly propagated. Improvement recommendation: add an automated check comparing the vacancy indicator to a related administrative series. Recommendations fed back to design.

#### Scenario B: Structural Business Statistics Composite Product

**Design Phase:** The composite SBS product is specified to integrate the register-based and survey-based turnover/employment estimates for the same NACE × region domains, with internal-consistency checks (domains sum to totals) and coherence vs. national-accounts inputs.

**Implementation Phase:** Engineers configure integration of the pinned `OD-BUS` outputs, the consistency checks, and dominance-aware confidentiality validation for the composite. Coherence and reproducibility tests pass.

**Execution Phase:** Q3-2026: an Output Data Registered event arrives. The register- and survey-based components are integrated for overlapping domains; NACE-class turnover sums to the published total; the components agree within tolerance except one domain, which is returned to VS-04 for review rather than adjusted. Once resolved, the product is approved and registered (`SP-BUS-2026Q3-001`) with lineage to its components; VS-06.0 is notified.

**Management Phase:** One coherence discrepancy this cycle (resolved upstream); component reuse rate high; reproducibility audit passes. Improvement recommendation: tighten the tolerance for the recurring domain and document the reconciliation. Recommendations fed back to design.

#### Scenario C: Consumer Price Index Product

**Design Phase:** The CPI product is specified to compose the headline all-items index and main divisions from the validated item indices, with chaining/rebasing rules, coherence checks against the prior month and year, and analytical commentary.

**Implementation Phase:** Engineers configure the index-composition engine (designed weights and chaining), coherence checks, and the commentary/approval tooling. Coherence and reproducibility tests pass; the composed headline matches the expected value on test data.

**Execution Phase:** For the monthly run, an Output Data Registered event arrives with the validated item indices. The headline and division indices are composed with the period weights, coherence checks confirm consistency with the prior month and year, and the commentary explains the main contributors to the monthly change. The product is approved and registered with full lineage; VS-06.0 is notified.

**Management Phase:** Coherence pass rate 100%, headline reproducible from item indices + weights, approval in one iteration. Improvement recommendation: enhance the contributions visualization for the commentary. Recommendations fed back to design.